

# Horn and can assembly

Construction guide | 2021 design

Build a horn from foil-faced insulation board and attach it to a metal can receiver. A separate cradle supports the assembly and lets you adjust its elevation.

## Materials

| Part              | Specification                                        |
|-------------------|------------------------------------------------------|
| Insulation board  | 4 × 8-foot sheet of foil-faced insulation board      |
| Metal can         | New, empty 1-gallon F-style metal container          |
| Copper probe      | 4- or 6-gauge copper wire, 5.25 cm long              |
| Connector         | Four-hole, panel-mount male bulkhead SMA feedthrough |
| Aluminum flashing | Two pieces, each 2 × 6 inches                        |
| Tape              | Aluminum foil tape and duct tape                     |

One insulation sheet makes one horn; two sheets can make three horns. Foil tape covers the interior seams and protects the exposed edges.

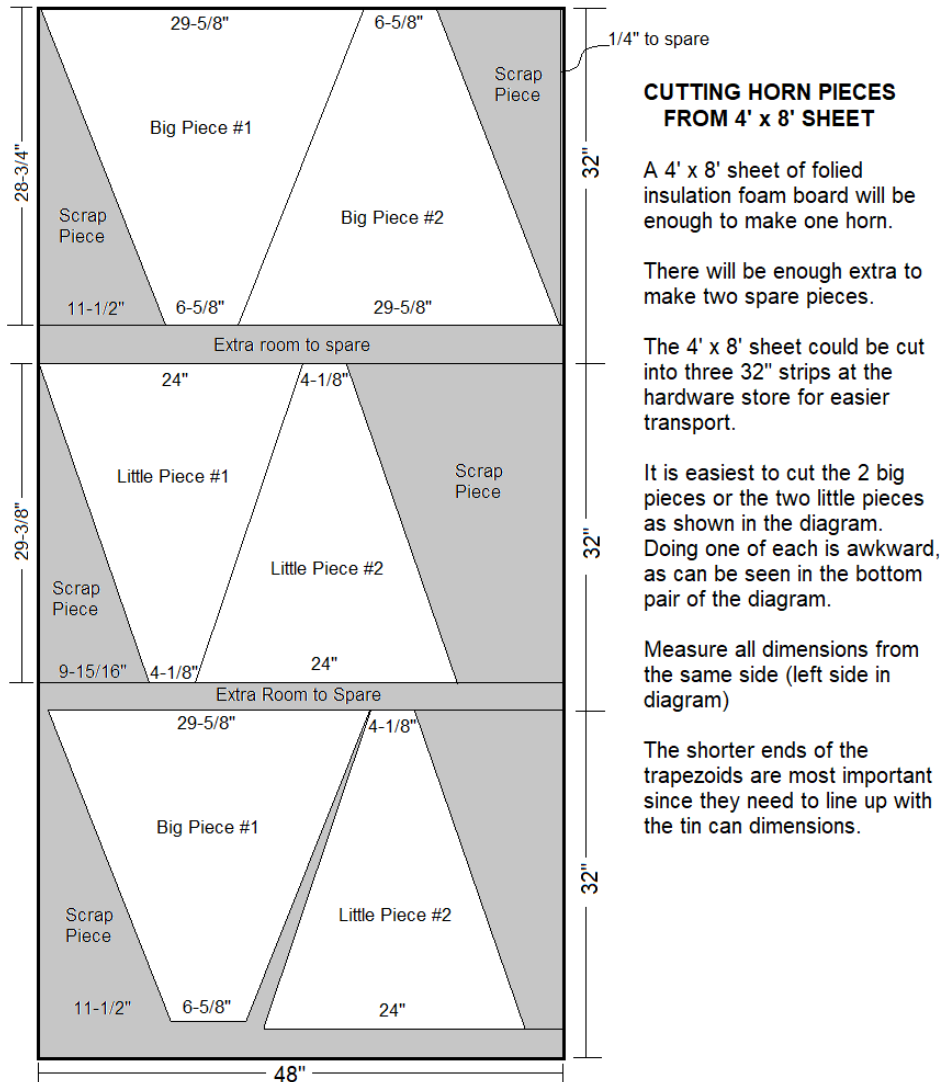
Use a new, empty can. Do not cut, drill, or heat a container that held flammable liquid.



**Figure 1. An assembled foil-faced horn.**

## Cut the horn panels

Cut two large panels and two small panels using the layout in Figure 2. Keep the shorter edges accurate so they fit the can opening.



## Join the horn panels

Lay the panels flat with their foil faces upward. Alternate large and small panels, matching the short edges carefully.

Tape the adjoining edges with foil tape. Fold the panels into a horn, keeping the foil faces inside. Tape the remaining seam, then seal the top and bottom edges.

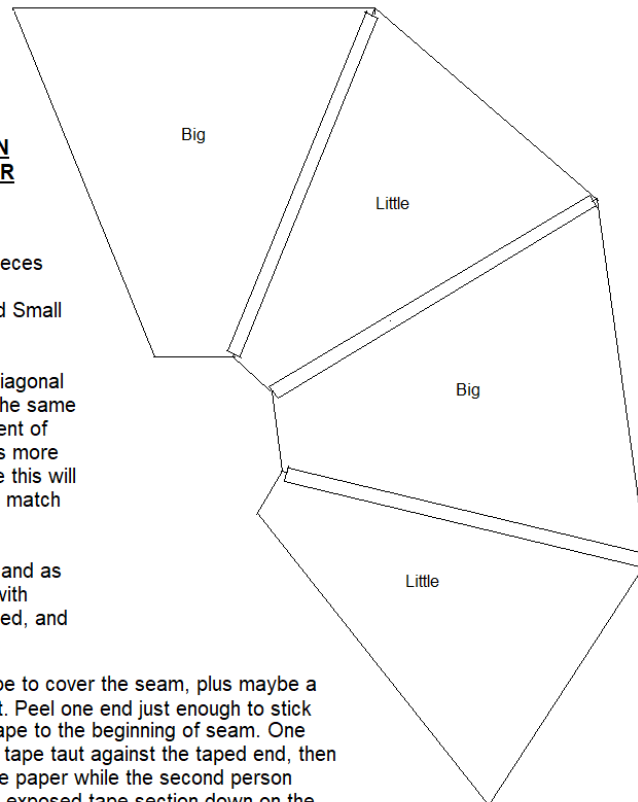
### TAPING THE HORN PIECES TOGETHER

Line up the four pieces on a flat surface alternating Big and Small pieces.

Even though the diagonal edges should be the same length, the alignment of the shorter ends is more important because this will be put together to match up with the can.

So, line up edges and as best as possible with smaller ends aligned, and get ready to tape.

Roll out the foil tape to cover the seam, plus maybe a 1/2 inch extra. Cut. Peel one end just enough to stick down the end of tape to the beginning of seam. One person should pull tape taut against the taped end, then start to peel off the paper while the second person presses the newly exposed tape section down on the seam.



**Figure 3. Panel order and seam taping before folding the horn.**

## Prepare the can receiver

1. Remove the bottom of the can with a smooth-edge can opener.
2. Solder the 5.25 cm copper probe to the SMA feedthrough pin.
3. Mark the midpoint of the can side, 5.25 cm from the capped end. Drill a 1/4-inch hole at this point.
4. Place a wood block inside the can while drilling to prevent the wall from bending. Remove the block afterward.
5. Solder the SMA feedthrough to the can with the copper probe extending inside. Keep the probe clear of the metal wall.



**Figure 4. Probe, drilled can, and attached feedthrough, from left to right.**

The photos show the connector outside the can and the probe extending into the receiver.

## Attach the can to the horn

1. Bend each 2 × 6-inch flashing piece into a squared S shape. The pieces support the joint between the can and horn.
2. Place the horn upside down. Tape the flashing pieces to the horn, as shown at left in Figure 5.
3. Align the open can with the horn opening. Keep the flashing outside the can.
4. Fit the flashing around the joint. Secure it to the can with duct tape or foil tape.
5. Cover the inside seams with foil tape. Close all gaps between the horn and can.



Figure 5. Flashing before and after attaching the can.



Figure 6. Interior seams fully covered with foil tape.